SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1707**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I V.Lasya(192524186) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: v.lasya

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. x: Vehicle(x) EmergencyType(x) → ClearPath(x) [Rule 1]

II. ∀∀∀x: ∀ClearPath(x)∧ →∧ GreenSignal(x)∧ ∧ ¬RedSignal(x) [Rule 2]

III. x y: Vehicle(x) Behind(x,y) GreenSignal(y) → Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: SoilDry → IrrigationNeeded
2) P2: IrrigationNeeded CropWheat → ApplyDripMethod
3) P3: ¬ApplyDripMethod∧ ∧ CropWheat → CropAtRisk
4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1) Stench [1,2] → Wumpus is adjacent to cell [1,2]
2) Breeze [1,1] → Pit is adjacent to cell [1,1]
3) Glitter [x,y] → Gold is at [x,y]
4) ¬Wumpus [x,y]∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts

Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandabl e presentation	Limited clarity and structure	Poor clarity; difficult to understand

Case Study 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

F = Patient A has Fever

C = Patient A has Cough

B = Patient A has Breathlessness

Flu = Patient A has Possible Flu

M = Patient A has Possible Measles

P = Patient A has Possible Pneumonia

Rule 1: $F \land C \rightarrow Flu$

Rule 2: $F \land Rash \rightarrow M$

Rule 3: $C \land B \rightarrow P$

Facts:

F = True

C = True

B = True

(b) Forward Chaining

Initial Facts: Fever, Cough, Breathlessness

Step 1:

Fever ∧ Cough → Possible Flu

Therefore, Possible Flu is derived.

Step 2:

Fever ∧ Rash → Possible Measles

Rash is not given. Therefore, Possible Measles cannot be derived.

Step 3:

Cough ∧ Breathlessness → Possible Pneumonia

Both conditions are true. Therefore, Possible Pneumonia is derived.

Final Conclusions:

Fever, Cough, Breathlessness, Possible Flu, Possible Pneumonia.

(c) Backward Chaining

Goal: Possible Pneumonia

Relevant rule:

Cough ∧ Breathlessness → Possible Pneumonia

```
          Goal reduction:
        Possible Pneumonia
                ↓
   Cough AND Breathlessness
              /    \
      Cough    Breathlessness
        True        True
              \    /
           Goal Proven
```

Therefore, Patient A has Possible Pneumonia is proven.

Case Study 2 — Autonomous Traffic Management Agent

(a) Unification

Rule 1:

∀x: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)

Given Facts:

Vehicle(Ambulance)

EmergencyType(Ambulance)

Match:

Vehicle(x) with Vehicle(Ambulance)

θ_1 = {x/Ambulance}

Match:

EmergencyType(x) with EmergencyType(Ambulance)

θ_2 = {x/Ambulance}

Therefore, the MGU is:

θ = {x/Ambulance}

After substitution:

Vehicle(Ambulance) ∧ EmergencyType(Ambulance) → ClearPath(Ambulance)

Therefore, ClearPath(Ambulance) is derived.

(b) Resolution Method

Rule 1 CNF:

¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)

Rule 2 CNF:

¬ClearPath(x) ∨ GreenSignal(x)

Rule 3 CNF:

¬Vehicle(x) ∨ ¬Behind(x,y) ∨ ¬GreenSignal(y) ∨ Proceed(x)

Facts:

Vehicle(Ambulance)

EmergencyType(Ambulance)

Vehicle(CarA)

Behind(CarA, Ambulance)

From Rule 1 and the ambulance facts:

ClearPath(Ambulance)

From Rule 2:

GreenSignal(Ambulance)

Using Rule 3:

Vehicle(CarA) ∧ Behind(CarA, Ambulance) ∧ GreenSignal(Ambulance)

Therefore:

Proceed(CarA)

Hence, Proceed(CarA) is proven.

(c) Two Simultaneous Emergency Vehicles

The current knowledge base is not completely sufficient for two simultaneous emergency vehicles.

Additional rules are required for:

- Emergency vehicle priority
- Route conflict resolution
- Intersection coordination
- Prevention of conflicting green signals

For example:

HigherPriority(x,y) ∧ EmergencyType(x) → PriorityPath(x)

PriorityPath(x) → GreenSignal(x)

Additional facts such as:

HigherPriority(Ambulance1, Ambulance2)

can also be included.

Therefore, priority and conflict-resolution rules are required.

Case Study 3 — Agricultural Expert Reasoning System

(a) CNF Conversion

P1

SoilDry → IrrigationNeeded

Eliminate implication:

¬SoilDry ∨ IrrigationNeeded

Therefore, CNF is:

¬SoilDry ∨ IrrigationNeeded

P2

IrrigationNeeded ∧ CropWheat → ApplyDripMethod

Eliminate implication:

¬(IrrigationNeeded ∧ CropWheat) ∨ ApplyDripMethod

Apply De Morgan's law:

¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod

Therefore, CNF is:

¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod

P3

¬ApplyDripMethod ∧ CropWheat → CropAtRisk

Eliminate implication:

¬(¬ApplyDripMethod ∧ CropWheat) ∨ CropAtRisk

Apply De Morgan's law:

ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk

Therefore, CNF is:

ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk

Facts

SoilDry = True

CNF:

SoilDry

CropWheat = True

CNF:

CropWheat

Final CNF

1. ¬SoilDry ∨ IrrigationNeeded
2. ¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod
3. ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk
4. SoilDry
5. CropWheat

(b) Resolution Refutation

Goal: ApplyDripMethod

Negate the goal:

¬ApplyDripMethod

From:

SoilDry

and

¬SoilDry ∨ IrrigationNeeded

we derive:

IrrigationNeeded

Using:

IrrigationNeeded

CropWheat

and

¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod

we derive:

ApplyDripMethod

Now resolve:

ApplyDripMethod

with:

¬ApplyDripMethod

Therefore:

□

The empty clause is obtained.

Hence, ApplyDripMethod is proven.

Resolution Proof Tree

SoilDry

|

↓

¬SoilDry ∨ IrrigationNeeded

|

↓

IrrigationNeeded

|

↓

¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod

|

↓

ApplyDripMethod

|

↓

¬ApplyDripMethod

(c) Removing SoilDry

If SoilDry is removed:

Irrigation Needed cannot be derived because Rule 1 requires SoilDry.

Without IrrigationNeeded:

Apply Drip Method cannot be derived.

For Crop At Risk:

¬ApplyDripMethod ∧ CropWheat → CropAtRisk

Although CropWheat is true, ¬ApplyDripMethod is not known.

Therefore:

CropAtRisk cannot be concluded.

Final Result

IrrigationNeeded → Not derived

ApplyDripMethod → Not derived

CropAtRisk → Not derived

Case Study 4 — Wumpus World Knowledge Agent

(a) Belief State

Given Percepts

Stench[1,2]

Breeze[1,1]

Glitter[2,2]

Rule 1

Stench[1,2] → Wumpus is adjacent to [1,2]

Therefore, possible Wumpus locations are:

[1,1], [1,3], [2,2]

Rule 2

Breeze[1,1] → Pit is adjacent to [1,1]

Therefore, possible Pit locations are:

[1,2], [2,1]

Rule 3

Glitter[x,y] → Gold[x,y]

Since:

Glitter[2,2]

Therefore:

Gold[2,2]

Rule 4

¬Wumpus[x,y] ∧ ¬Pit[x,y] → Safe[x,y]

No negative Wumpus or Pit facts are provided. Therefore, no additional Safe cells can be conclusively derived.

(b) Forward Chaining

Wumpus

Stench[1,2]

↓

Wumpus is adjacent to [1,2]

↓

Possible locations:

[1,1], [1,3], [2,2]

Pit

Breeze[1,1]

↓

Pit is adjacent to [1,1]

↓

Possible locations:

[1,2], [2,1]

Gold

Glitter[2,2]

↓

Gold[2,2]

(c) Path Safety Analysis

Given path:

[1,1] → [1,2] → [2,2]

Cell [1,1]

Breeze at [1,1] indicates a possible pit in an adjacent cell.

Therefore, [1,1] is not proven dangerous, but it provides a warning about neighboring cells.

Cell [1,2]

From Breeze[1,1]:

[1,2] is a possible Pit location.

Therefore, [1,2] cannot be confirmed safe.

Cell [2,2]

From Stench[1,2]:

[2,2] is a possible Wumpus location.

Also:

Gold[2,2]

Therefore, although the gold is present, [2,2] cannot be confirmed safe.

Final Conclusion

The path:

[1,1] → [1,2] → [2,2]

is not guaranteed to be safe based on the current knowledge base. More information is required before the agent moves through these cells.